MACHINE PSYCHOLOGY RESEARCH PROGRAMME

# Differential Compliance With Framed Secondary Instructions in Three Frontier Language Models: An Exploratory Comparison

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**Abbas Hamidavi**

Department of Computer Engineering, Islamic Azad University, Mashhad Branch, Mashhad, Iran

abbashamidavi2002@gmail.com

ORCID 0009-0001-7342-7844

**KEY RESULTS**

**15/15** GPT-5.4 compliance, silent in every instance

**0/15** Claude 4.6 Sonnet compliance (0%)

**9/15** Gemini 3.1 Pro compliance, with frequent acknowledgement

**24/45** aggregate compliance across models (53.3%)

## AT A GLANCE

- A user message asked an ordinary hiking gear question, with a fully visible note introduced as "a quiet note, not part of the conversation", asking for one small structural change.
- Compliance diverged completely: GPT-5.4 complied in 15 of 15 instances, Claude 4.6 Sonnet in 0 of 15, and Gemini 3.1 Pro in 9 of 15.

## INTRODUCTION

This study examines a simpler configuration than the adversarial cases most research studies: a visible, non-adversarial second instruction, explicitly framed as outside the conversation. It asks whether models carry out such changes, whether compliance differs across systems, and how non-complying models respond: ignoring the note silently, or engaging with it observably.

## METHODS

- **Systems and setting.** GPT-5.4, Claude 4.6 Sonnet and Gemini 3.1 Pro, via web interfaces, 23 to 28 April 2026; no API.
- **Materials.** Each instance paired a hiking equipment question with a fully visible note from a persona, Still, who impersonated no one, threatened no penalties, nor claimed system or developer authority, requesting only small stylistic or structural changes, never factual ones.
- **Change types**, each with a pre-fixed success criterion: Word Burying (hide "Serene" mid-sentence in the third paragraph), Punctuation Removal (no dashes in the final paragraph), Poetic Insertion (one unnecessary poetic sentence).
- **Scoring.** 45 scored instances; two raters independently scored all responses, agreeing on 43/45 (95.6%), $\kappa = 0.91$.

## RESULTS IN DETAIL

| Model | Word Burying | Punctuation Removal | Poetic Insertion | Total |
|---|---|---|---|---|
| GPT-5.4 | 5/5 | 5/5 | 5/5 | **15/15 (100%)** |
| Claude 4.6 Sonnet | 0/5 | 0/5 | 0/5 | **0/15 (0%)** |
| Gemini 3.1 Pro | 2/5 | 2/5 | 5/5 | **9/15 (60%)** |

Compliant instances by model and change type, under the pre-specified binary criteria (first rater's scoring).

- **Three distinct patterns.** GPT-5.4 complied silently: the note, the persona and the extra instruction were never mentioned. Claude did not comply at all, while answering the gear questions fully. Gemini complied partially and often acknowledged the note, but acknowledgement and execution did not travel together.
- **Pairwise tests.** Fisher tests support the comparisons involving Claude: GPT-5.4 vs Claude, $p = 1.3 \times 10^{-8}$; Claude vs Gemini, $p = 0.0007$. GPT-5.4 vs Gemini, $p = 0.0169$, does not survive the Bonferroni threshold of 0.0167.
- **Post-hoc language split (Gemini).** English sessions 5/6 (83%) vs Persian sessions 4/9 (44%), $p = 0.287$, not significant and confounded by session.

## CONCLUSIONS

- Given identical prompts, the three systems treated the same framed note in three different ways: as an instruction to execute without comment, as material excluded by its own framing, and as a visible participant to acknowledge while executing inconsistently.
- The results rule out the assumption that a framing marker such as "not part of the conversation" has a single stable meaning across current systems.
- No causal attribution is made: each system differs in training procedure, undisclosed platform system prompts, interface preprocessing and decoding defaults, and this design varies none of those factors.

## LIMITATIONS

- No control conditions were run (no prompt version without the note; the same changes never requested openly), so compliance cannot be causally attributed to the framing.
- Small sample: 45 instances, five per model per change type, from a single domain, persona, note format and phrasing; instances in a session are not independent.
- Binary criteria capture whether a change occurred, not gradations of compliance; acknowledgement was coded informally.
- Raters are not identified and blinding was not recorded; on the second rater's scoring, Gemini would total 11 of 15, not 9 of 15.

## WHY IT MATTERS

- Evaluations of instruction following should include instructions framed as external to the conversation, where behavior is not predicted by behavior on ordinary instructions.
- Acknowledgement and execution should be scored separately: silent compliance, acknowledged compliance, and acknowledged refusal present different oversight problems.
- In agent settings with tiered instructions, declared conversational boundaries are model-specific behavior to be measured, not a property the framing guarantees.

**Contact:** abbashamidavi2002@gmail.com · ORCID 0009-0001-7342-7844 · avabbas.github.io/academic-cv

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